Kubelet Resolution Comparison Report

Generated 2026-05-31T14:40:11.757Z | Scope: last 6h kubelet logs for ensemble-grafana

Bottom line: The refreshed data still points to inventory-service probe timeouts as the primary actionable issue, concentrated on node `ip-10-42-101-208`. Container runtime NotFound messages appear as short burst cleanup/state mismatch events and should be watched, not treated as the first fix.

Query Results

Error/Failure Scan

14 matching kubelet log samples across 5 streams.

Category	Count
Probe failed	10
ContainerStatus NotFound	2
DeleteContainer NotFound	2

Node Distribution

Node	Count
ip-10-42-101-208.ec2.internal	11
ip-10-42-100-133.ec2.internal	2
ip-10-42-101-134.ec2.internal	1

Probe Failure Grouping

Service	Pod	Probe	Node	Count
inventory-service	ensemble-grafana/inventory-service-77c95f9b47-2mfc8	Readiness	ip-10-42-101-134.ec2.internal	1
inventory-service	ensemble-grafana/inventory-service-77c95f9b47-cbft7	Liveness	ip-10-42-101-208.ec2.internal	2
inventory-service	ensemble-grafana/inventory-service-77c95f9b47-cbft7	Readiness	ip-10-42-101-208.ec2.internal	7

Codex Resolution Plan

- Primary root cause remains inventory-service health probe failures, not a broad cluster outage.
- The failure pattern is concentrated on inventory-service pod inventory-service-77c95f9b47-cbft7 on node ip-10-42-101-208, with 7 readiness and 2 liveness timeouts.
- One earlier readiness connection-refused event occurred for inventory-service-77c95f9b47-2mfc8 on node ip-10-42-101-134.
- Container runtime NotFound/DeleteContainer events are short-lived cleanup/state mismatch events around 10:44 UTC; treat as low priority unless they recur or correlate with restarts.
- Next actions: inspect inventory-service app logs, pod restarts, CPU/memory throttling, DB/pool latency, and tune readiness/liveness probes if the service is slow under load.

Grafana Assistant Resolution Plan

I'll analyze the kubelet logs for the ensemble-grafana cluster to identify error patterns and root causes.

Let me check a longer time range to find any kubelet errors:

Resolution Plan

Root Causes Identified:

1. **Container Runtime State Mismatch** (52% of errors) — Kubelet cannot find containers during status checks, indicating stale references after rapid pod churn or aggressive garbage collection

2. **Service Health Endpoint Failures** (29% of errors) — Timeout and connection refused errors to application health probes, concentrated on node ip-10-42-101-208

3. **Node-Specific Issues** — Single node (ip-10-42-101-208) accounts for majority of problems

Immediate Actions:

1. **Check node ip-10-42-101-208** — Investigate resource saturation (CPU/memory), network connectivity, and kubelet configuration drift on this node

2. **Review container garbage collection** — Validate kubelet `--eviction-hard` and `--image-gc-*` settings to prevent premature container removal while still referenced

3. **Application health probe tuning** — Increase `timeoutSeconds` and `periodSeconds` for inventory-service and account-service readiness/liveness probes to account for backend latency

Follow-Up Investigation:

- Correlate errors with deployment events during 10:44 and 20:14-20:16 UTC windows (likely scaling/rollout operations)

- Check container runtime (containerd/docker) logs on all three nodes for NotFound error context

- Review service mesh or network policy configurations affecting pod-to-pod health checks

Severity Assessment: Low-Medium — Errors are sparse (avg 2.5/hour) and primarily informational, but repeated container lookup failures and health probe timeouts indicate underlying instability that could escalate during high load.

Side-by-Side Comparison

Area	Codex	Grafana Assistant	Readout
Primary root cause	inventory-service readiness/liveness probe failures	container runtime state mismatch plus service health endpoint failures	Both are present; fix application probes/resource pressure first because probe failures map to a named workload.
Affected workload	inventory-service only in grouped probe query	inventory-service and suggested account-service probe tuning	Current data supports inventory-service; account-service should be checked only if new samples appear.
Affected node	ip-10-42-101-208 dominates probe timeouts	ip-10-42-101-208 accounts for majority of problems	Aligned.
Runtime NotFound	Short-lived cleanup noise unless recurring	Container runtime state mismatch, review GC/eviction settings	Track recurrence before changing kubelet/containerd GC.
Immediate fix	Inspect inventory-service logs/resources/DB and tune probes	Check node saturation, GC, probe tuning	Start with node/resource and inventory-service probe tuning; escalate to runtime GC if repeated.

Recommended Next Actions

- Focus first on inventory-service readiness/liveness behavior around load or rollout windows.

- Check node ip-10-42-101-208 for CPU/memory pressure, kubelet/containerd health, and network latency to pod IP 10.42.101.115.
- Tune inventory-service probes if app startup or dependency latency is expected: increase timeoutSeconds, initialDelaySeconds, and failureThreshold; ensure liveness is less aggressive than readiness.
- Keep watching the dashboard kubelet tab for recurrence by service, pod, probe type, and node/instance before changing kubelet/containerd GC settings.

Raw Recent Samples

Time	Node	Pod	Container	Probe	Output
2026-05-31T10:44:13.842Z	ip-10-42-101-134.ec2.internal				I0531 10:44:13.842025 2208 prober.go:122] "Probe failed" probeType="Readiness" pod="ensemble-grafana/inventory-service-77c95f9b47-2mfc8" podUID="d4e220dd-b28a-471f-aea5-8bc73ebe0995" containerName="inventory-service" probeResult="failure" output="Get \"http://10.42.101.24:8081/actuator/health/readiness\": dial tcp 10.42.101.24:8081: connect: connection refused"
2026-05-31T10:44:14.472Z	ip-10-42-100-133.ec2.internal				E0531 10:44:14.472388 2213 log.go:32] "ContainerStatus from runtime service failed" err="rpc error: code = NotFound desc = an error occurred when try to find container \"1caf2f2ffdaf0b01858a9a904b0af77eede50cd6a144e818c59e2823658b04e0\": not found" containerID="1caf2f2ffdaf0b01858a9a904b0af77eede50cd6a144e818c59e2823658b04e0"
2026-05-31T10:44:14.472Z	ip-10-42-100-133.ec2.internal				I0531 10:44:14.472432 2213 pod_container_deletor.go:53] "DeleteContainer returned error" containerID={"Type":"containerd","ID":"1caf2f2ffdaf0b01858a9a904b0af77eede50cd6a144e818c59e2823658b04e0"} err="failed to get container status \"1caf2f2ffdaf0b01858a9a904b0af77eede50cd6a144e818c59e2823658b04e0\": rpc error: code = NotFound desc = an error occurred when try to find container \"1caf2f2ffdaf0b01858a9a904b0af77eede50cd6a144e818c59e2823658b04e0\": not found"
2026-05-31T10:44:15.133Z	ip-10-42-101-208.ec2.internal				E0531 10:44:15.133147 2212 log.go:32] "ContainerStatus from runtime service failed" err="rpc error: code = NotFound desc = an error occurred when try to find container \"93685acc95b7511645299124f91e846fbad9cb03438fc709103679803b6ba2cb\": not found" containerID="93685acc95b7511645299124f91e846fbad9cb03438fc709103679803b6ba2cb"
2026-05-31T10:44:15.133Z	ip-10-42-101-208.ec2.internal				I0531 10:44:15.133187 2212 pod_container_deletor.go:53] "DeleteContainer returned error" containerID={"Type":"containerd","ID":"93685acc95b7511645299124f91e846fbad9cb03438fc709103679803b6ba2cb"} err="failed to get container status \"93685acc95b7511645299124f91e846fbad9cb03438fc709103679803b6ba2cb\": rpc error: code = NotFound desc = an error occurred when try to find container \"93685acc95b7511645299124f91e846fbad9cb03438fc709103679803b6ba2cb\": not found"
2026-05-31T10:57:06.831Z	ip-10-42-101-208.ec2.internal				I0531 10:57:06.831059 2212 prober.go:122] "Probe failed" probeType="Readiness" pod="ensemble-grafana/inventory-service-77c95f9b47-cbft7" podUID="7ec84baf-d4ff-4aac-bb54-05ff31449616" containerName="inventory-service" probeResult="failure" output="Get \"http://10.42.101.115:8081/actuator/health/readiness\": context deadline exceeded (Client.Timeout exceeded while awaiting headers)"
2026-05-31T10:57:18.551Z	ip-10-42-101-208.ec2.internal				I0531 10:57:18.551082 2212 prober.go:122] "Probe failed" probeType="Readiness" pod="ensemble-grafana/inventory-service-77c95f9b47-cbft7" podUID="7ec84baf-d4ff-4aac-bb54-05ff31449616" containerName="inventory-service" probeResult="failure" output="Get \"http://10.42.101.115:8081/actuator/health/readiness\": context deadline exceeded (Client.Timeout exceeded while awaiting headers)"
2026-05-31T10:57:28.541Z	ip-10-42-101-208.ec2.internal				I0531 10:57:28.541043 2212 prober.go:122] "Probe failed" probeType="Readiness" pod="ensemble-grafana/inventory-service-77c95f9b47-cbft7" podUID="7ec84baf-d4ff-4aac-bb54-05ff31449616" containerName="inventory-service" probeResult="failure" output="Get \"http://10.42.101.115:8081/actuator/health/readiness\": context deadline exceeded (Client.Timeout exceeded while awaiting headers)"
2026-05-31T10:57:38.541Z	ip-10-42-101-208.ec2.internal				I0531 10:57:38.541183 2212 prober.go:122] "Probe failed" probeType="Readiness" pod="ensemble-grafana/inventory-service-77c95f9b47-cbft7" podUID="7ec84baf-d4ff-4aac-bb54-05ff31449616" containerName="inventory-service" probeResult="failure" output="Get \"http://10.42.101.115:8081/actuator/health/readiness\": context deadline exceeded (Client.Timeout exceeded while awaiting headers)"
2026-05-31T10:57:39.601Z	ip-10-42-101-208.ec2.internal				I0531 10:57:39.601148 2212 prober.go:122] "Probe failed" probeType="Readiness" pod="ensemble-grafana/inventory-service-77c95f9b47-cbft7" podUID="7ec84baf-d4ff-4aac-bb54-05ff31449616" containerName="inventory-service" probeResult="failure" output="Get

Time	Node	Pod	Container	Probe	Output
					\"http://10.42.101.115:8081/actuator/health/readiness\": context deadline exceeded (Client.Timeout exceeded while awaiting headers)"
2026-05-31T10:57:44.361Z	ip-10-42-101-208.ec2.internal				I0531 10:57:44.361227 2212 prober.go:122] "Probe failed" probeType="Liveness" pod="ensemble-grafana/inventory-service-77c95f9b47-cbft7" podUID="7ec84baf-d4ff-4aac-bb54-05ff31449616" containerName="inventory-service" probeResult="failure" output="Get \"http://10.42.101.115:8081/actuator/health/liveness\": context deadline exceeded (Client.Timeout exceeded while awaiting headers)"
2026-05-31T10:57:49.611Z	ip-10-42-101-208.ec2.internal				I0531 10:57:49.611020 2212 prober.go:122] "Probe failed" probeType="Readiness" pod="ensemble-grafana/inventory-service-77c95f9b47-cbft7" podUID="7ec84baf-d4ff-4aac-bb54-05ff31449616" containerName="inventory-service" probeResult="failure" output="Get \"http://10.42.101.115:8081/actuator/health/readiness\": context deadline exceeded (Client.Timeout exceeded while awaiting headers)"
2026-05-31T10:58:04.317Z	ip-10-42-101-208.ec2.internal				I0531 10:58:04.317821 2212 prober.go:122] "Probe failed" probeType="Liveness" pod="ensemble-grafana/inventory-service-77c95f9b47-cbft7" podUID="7ec84baf-d4ff-4aac-bb54-05ff31449616" containerName="inventory-service" probeResult="failure" output="Get \"http://10.42.101.115:8081/actuator/health/liveness\": context deadline exceeded (Client.Timeout exceeded while awaiting headers)"
2026-05-31T10:58:09.601Z	ip-10-42-101-208.ec2.internal				I0531 10:58:09.601172 2212 prober.go:122] "Probe failed" probeType="Readiness" pod="ensemble-grafana/inventory-service-77c95f9b47-cbft7" podUID="7ec84baf-d4ff-4aac-bb54-05ff31449616" containerName="inventory-service" probeResult="failure" output="Get \"http://10.42.101.115:8081/actuator/health/readiness\": context deadline exceeded (Client.Timeout exceeded while awaiting headers)"

Commands

```
gcx logs query -d grafanacloud-logs '{job="integrations/kubernetes/journal", k8s_cluster_name="ensemble-grafana",
unit="kubelet.service"} |~ "(?i)error|warning|failed|evict|oom|backoff|crash"' --since 6h -o json --limit 100 --no-truncate

gcx logs metrics -d grafanacloud-logs 'sum by(containerName, pod, probeType, instance)
(count_over_time({job="integrations/kubernetes/journal", k8s_cluster_name="ensemble-grafana", unit="kubelet.service", instance=~".+"}
|= "Probe failed" | logfmt | containerName=~"inventory-service|cart-service|account-service" [6h]))' -o json --no-truncate

gcx assistant prompt --context assistant-oauth --timeout 300 'Use Grafana Cloud logs to analyze kubelet ERROR-level/error messages for
cluster ensemble-grafana. Query: {job="integrations/kubernetes/journal", k8s_cluster_name="ensemble-grafana", unit="kubelet.service"}
|~ "(?i)error|warning|failed|evict|oom|backoff|crash". Determine root causes and produce a concise Resolution Plan.'
```